American River College – Math 372
Spring 2020 – Class 26/32

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Class:	A & S 102, TTh 8:15 – 10:20
Student hours:	Howard Hall 148, MW 1:00 – 2:30, TTh 1:00 – 2:00

Today:	§§ 11.5/11.6
Assignment:	Online Homework #12
Next:	§§ 12.1/12.2

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11.5 Rotation of Axes

So far the parabolas, ellipses and hyperbolas we've studied have all had their axes of symmetry parallel to the x- and/or y-axis. Their equations were always of the form

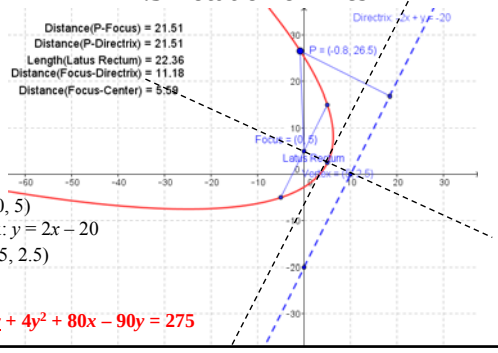
$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

What about parabolas, ellipses and hyperbolas that do not have their axes of symmetry parallel to either the x- or y-axis? What do their equations look like?

Consider a parabola with focus at (0, 5) with directrix $y = 2x - 20$

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11.5 Rotation of Axes



Focus: (0, 5)
Directrix: $y = 2x - 20$
Vertex: (5, 2.5)

$$x^2 + 4xy + 4y^2 + 80x - 90y = 275$$

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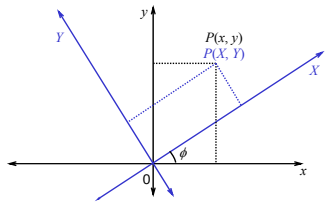
11.5 Rotation of Axes

In general the graph of $Ax^2 + \mathbf{Bxy} + Cy^2 + Dx + Ey + F = 0$ is also a conic. In fact, we can eliminate the "xy" term by rotating the axes through an appropriate angle. In the new, rotated coordinate system, the equation we obtain can be analyzed and sketched as in the previous sections.

Our task is to determine this appropriate angle and the resulting equation in the new coordinate system.

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11.5 Rotation of Axes

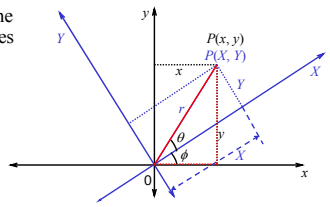


Rotate the x- and y-axes through the acute angle ϕ to produce the X- and Y-axes. Point P has coordinates (x, y) in the old system and coordinates (X, Y) in the new system.

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11.5 Rotation of Axes

Let r denote the distance of P from the origin and let θ be the angle OP makes with the new X-axis.



(x, y) and (X, Y) are related by

$$X = r\cos\theta \qquad Y = r\sin\theta \text{ and}$$
$$x = r\cos(\theta + \phi) \qquad y = r\sin(\theta + \phi)$$

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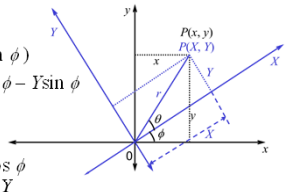
11.5 Rotation of Axes

Consider first $x = r\cos(\theta + \phi)$:

$x = r\cos(\theta + \phi) = r(\cos\theta\cos\phi - \sin\theta\sin\phi)$

$= (r\cos\theta)\cos\phi - (r\sin\theta)\sin\phi = X\cos\phi - Y\sin\phi$

Similarly, $y = X\sin\phi + Y\cos\phi$



Using $x = X\cos\phi - Y\sin\phi$ and $y = X\sin\phi + Y\cos\phi$ as a system of equations and solving for X and Y yields: $X = x\cos\phi + y\sin\phi$ and $Y = -x\sin\phi + y\cos\phi$

Finally, $x = X\cos\phi - Y\sin\phi$ $\quad X = x\cos\phi + y\sin\phi$

$y = X\sin\phi + Y\cos\phi$ $\quad Y = -x\sin\phi + y\cos\phi$

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11.5 Rotation of Axes

$x = X\cos\phi - Y\sin\phi$ $\quad X = x\cos\phi + y\sin\phi$

$y = X\sin\phi + Y\cos\phi$ $\quad Y = -x\sin\phi + y\cos\phi$

Example: If the coordinate axes are rotated through 30°, find the XY-coordinates of the point with xy-coordinates $(2, -4)$.

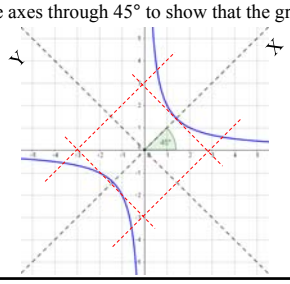
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11.5 Rotation of Axes

$x = X\cos\phi - Y\sin\phi$ $\quad X = x\cos\phi + y\sin\phi$

$y = X\sin\phi + Y\cos\phi$ $\quad Y = -x\sin\phi + y\cos\phi$

Example: Rotate the coordinate axes through 45° to show that the graph of the equation $xy = 2$ is a hyperbola.



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11.5 Rotation of Axes

Theorem: To eliminate the xy-term in the general conic equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

rotate the axes through the acute angle ϕ that satisfies $\cot 2\phi = \dfrac{A - C}{B}$

Proof: Substitute $x = X\cos\phi - Y\sin\phi$ and $y = X\sin\phi + Y\cos\phi$ above to get

$$\begin{aligned} A(X\cos\phi - Y\sin\phi)^2 &+ B(X\cos\phi - Y\sin\phi)(X\sin\phi + Y\cos\phi) \\ &+ C(X\sin\phi + Y\cos\phi)^2 + D(X\cos\phi - Y\sin\phi) \\ &+ E(X\sin\phi + Y\cos\phi) + F = 0 \end{aligned}$$

Expand and collect like terms to get

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11.5 Rotation of Axes

Theorem: To eliminate the xy-term in the general conic equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

rotate the axes through the acute angle ϕ that satisfies $\cot 2\phi = \dfrac{A - C}{B}$

Proof:

$$A'X^2 + B'XY + C'Y^2 + D'X + E'Y + F' = 0$$

where

$$\begin{aligned} A' &= A\cos^2\phi + B\sin\phi\cos\phi + C\sin^2\phi \\ B' &= 2(C - A)\sin\phi\cos\phi + B(\cos^2\phi - \sin^2\phi) \\ C' &= A\sin^2\phi - B\sin\phi\cos\phi + C\cos^2\phi \\ D' &= D\cos\phi + E\sin\phi \\ E' &= -D\sin\phi + E\cos\phi \\ F' &= F \end{aligned}$$

To eliminate the XY-term, we need to choose ϕ so that $B' = 0$.

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11.5 Rotation of Axes

Theorem: To eliminate the xy-term in the general conic equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

rotate the axes through the acute angle ϕ that satisfies $\cot 2\phi = \dfrac{A - C}{B}$

Proof: To eliminate the XY-term, we would like to choose ϕ so that $B' = 0$, that is,

$$2(C - A)\sin\phi\cos\phi + B(\cos^2\phi - \sin^2\phi) = 0$$

$$(C - A)\sin 2\phi + B\cos 2\phi = 0 \qquad \text{Double-Angle Formulas for Sine and Cosine}$$

$$B\cos 2\phi = (A - C)\sin 2\phi$$

$$\cot 2\phi = \frac{A - C}{B} \qquad \text{Divide by } B\sin 2\phi$$

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11.5 Rotation of Axes

Example: Use rotation of axes to eliminate the *xy*-term in the equation below, identify the conic and sketch

$\left(6\sqrt{3}\right)x^2 + 6xy + \left(4\sqrt{3}\right)y^2 = 21\sqrt{3}$ $\qquad Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$

$$\cot 2\phi = \frac{A-C}{B} = \frac{6\sqrt{3}-4\sqrt{3}}{6} = \frac{\sqrt{3}}{3}$$

$$\Rightarrow 2\phi = 60° = \frac{\pi}{3} \Rightarrow \phi = 30° = \frac{\pi}{6}$$

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11.5 Rotation of Axes

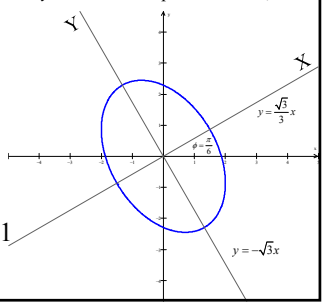
Example: Use rotation of axes to eliminate the *xy*-term in the equation below, identify the conic and sketch

$\left(6\sqrt{3}\right)x^2 + 6xy + \left(4\sqrt{3}\right)y^2 = 21\sqrt{3}$

$$\cot 2\phi = \frac{A-C}{B} = \frac{6\sqrt{3}-4\sqrt{3}}{6} = \frac{\sqrt{3}}{3}$$

$$\Rightarrow 2\phi = 60° = \frac{\pi}{3} \Rightarrow \phi = 30° = \frac{\pi}{6}$$

$$\left.\begin{aligned} \Rightarrow x &= X\left(\frac{\sqrt{3}}{2}\right) - Y\left(\frac{1}{2}\right) \\ \Rightarrow y &= X\left(\frac{1}{2}\right) + Y\left(\frac{\sqrt{3}}{2}\right) \end{aligned}\right\} \quad \frac{X^2}{3} + \frac{Y^2}{7} = 1$$



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11.5 Rotation of Axes

The following theorem eliminates the need to rotate the axes to identify a conic.

Theorem: The graph of the equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ is either a conic or a degenerate conic. In the non-degenerate cases, the graph is

1. A parabola if $B^2 - 4AC = 0$
2. An ellipse if $B^2 - 4AC < 0$
3. A hyperbola if $B^2 - 4AC > 0$

where $B^2 - 4AC$ is called the **discriminant**.

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11.6 Polar Equations of Conics

Parabolas have been defined in terms of a focus and directrix. However, ellipses and hyperbolas have been defined in terms of two foci.

We can define all three in terms of a focus and directrix by appealing to polar equations; in fact each conic (parabola, ellipse and hyperbola) can be defined as a simple polar equation.

Rotating conics is much simpler when using polar equations because the graph of $r = f(\theta - \alpha)$ is the graph of $r = f(\theta)$ rotated counterclockwise about the origin through an angle α.

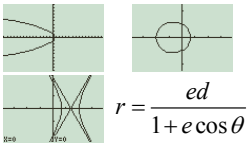
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11.6 Polar Equations of Conics

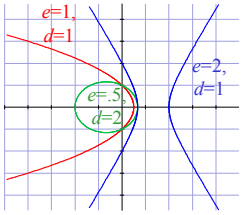
A polar equation of the form $r = \dfrac{ed}{1 \pm e\cos\theta}$ or $r = \dfrac{ed}{1 \pm e\sin\theta}$

represents a conic ***with one focus at the origin*** and with eccentricity e. The conic is:

1. A parabola if $e = 1$.
2. An ellipse if $0 < e < 1$.
3. A hyperbola if $e > 1$.



$$r = \frac{ed}{1 + e\cos\theta}$$



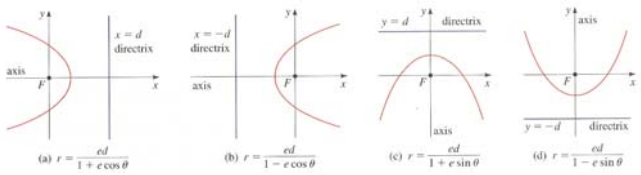
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11.6 Polar Equations of Conics

A polar equation of the form $r = \dfrac{ed}{1 \pm e\cos\theta}$ or $r = \dfrac{ed}{1 \pm e\sin\theta}$

represents a conic ***with one focus at the origin*** and with eccentricity e.

The form of the polar equation indicates the location of the directrix.

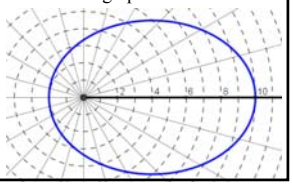


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11.6 Polar Equations of Conics

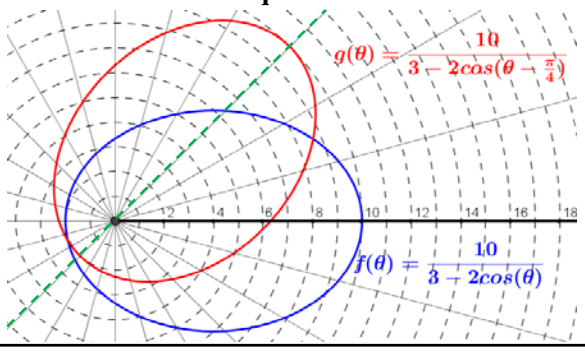
Consider the following polar equation: $r = \dfrac{10}{3 - 2\cos\theta}$

a. Show that the conic is an ellipse, and sketch the graph.
b. Find the center of the ellipse and the lengths of the major and minor axes.
c. Rotate the polar equation above through an angle $\pi/4$ about the origin. Find a polar equation for the resulting ellipse, and draw its graph.



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11.6 Polar Equations of Conics



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